	Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current	579 - distribution_elementwise_grid_stride_kernel	(144, 1, 1)x(256, 1, 1)	43.71 us	83,047	0 - NVIDIA GeForce RTX 4060 Laptop GPU	1.89 Ghz	[29504] python3.12	

Summary | Details | Source | Context | Comments | Raw | Session

Compare | Tools | View | Export

GPU Speed Of Light Throughput

GPU Throughput Chart

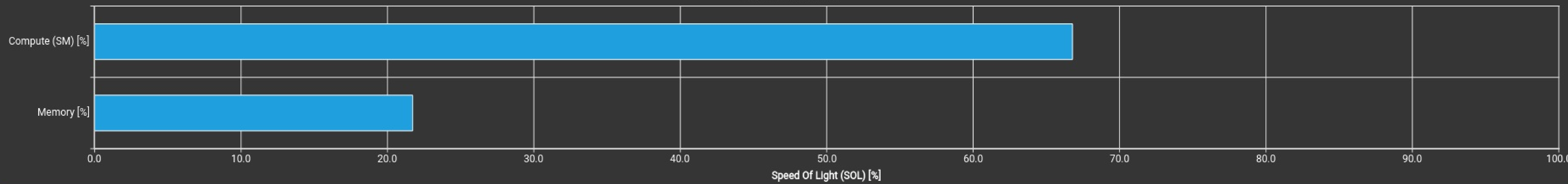
High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor.

Compute (SM) Throughput [%]	66.79	Duration [us]	43.71
Memory Throughput [%]	21.72	Elapsed Cycles [cycle]	83,047
L1/TEX Cache Throughput [%]	26.33	SM Active Cycles [cycle]	80,558.46
L2 Cache Throughput [%]	21.72	SM Frequency [Ghz]	1.89
DRAM Throughput [%]	0.08	DRAM Frequency [Ghz]	7.99

High Compute Throughput Compute is more heavily utilized than Memory: Look at the Compute Workload Analysis section to see what the compute pipelines are spending their time doing. Also, consider whether any computation is redundant and could be reduced or moved to look-up tables.

Key Performance Indicators

GPU Throughput



Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	144	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	50	Static Shared Memory Per Block [byte/block]	0
Block Size	256	Dynamic Shared Memory Per Block [byte/block]	0
Threads [thread]	36,864	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	1.50	Shared Memory Configuration Size [Kbyte]	16.38
Uses Green Context	0	Stack Size	1,024
# SMs [SM]	24	# TPCs	12
Enabled TPC IDs	all	-	-

Tail Effect
Est. Speedup: 50.00% A wave of thread blocks is defined as the maximum number of blocks that can be executed in parallel on the target GPU. The number of blocks in a wave depends on the number of multiprocessors and the theoretical occupancy of the kernel. This kernel launch results in 1 full waves and a partial wave of 48 thread blocks. Under the assumption of a uniform execution duration of all thread blocks, this partial wave may account for up to 50.0% of the total runtime of this kernel. Try launching a grid with no partial wave. The overall impact of this tail effect also lessens with the number of full waves executed for a grid. See the Hardware Model description for more details on launch configurations.

Key Performance Indicators

Occupancy

% Occupancy Graphs

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	66.67	Block Limit Registers [block]	4
Theoretical Active Warps per SM [warp]	32	Block Limit Shared Mem [block]	16
Achieved Occupancy [%]	54.65	Block Limit Warps [block]	6
Achieved Active Warps Per SM [warp]	26.23	Block Limit SM [block]	24

Achieved Occupancy
Est. Local Speedup: 18.03% The difference between calculated theoretical (66.7%) and measured achieved occupancy (54.6%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the CUDA Best Practices Guide for more details on optimizing occupancy.

Key Performance Indicators

Theoretical Occupancy
Est. Local Speedup: 33.33% The 8.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 12. This kernel's theoretical occupancy (66.7%) is limited by the number of required registers.

Key Performance Indicators

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	80,558.46	Average L1 Active Cycles [cycle]	80,558.46
Average L2 Active Cycles [cycle]	64,016.88	Average SMSP Active Cycles [cycle]	80,535.96
Average DRAM Active Cycles [cycle]	268	Average MC Channel Active Cycles [cycle] ⚠	n/a
Total SM Elapsed Cycles [cycle]	1,991,712	Total L1 Elapsed Cycles [cycle]	1,991,712
Total L2 Elapsed Cycles [cycle]	1,210,240	Total SMSP Elapsed Cycles [cycle]	7,966,848
Total DRAM Elapsed Cycles [cycle]	1,396,736	Total MC Channel Elapsed Cycles [cycle] ⚠	n/a

Missing Roofline and Memory Charts? Profile again with the *detailed* metric set to collect all necessary metrics.
Also consider collecting all available sections with the *full* metric set.